

```

1  #include <ESP8266WiFi.h>
2  #include <ESP8266HTTPClient.h>
3  #include <WiFiClient.h>
4  #include <Arduino_JSON.h>
5  #include <utility>
6
7  using namespace std;
8
9  const char* ssid = "WIFI_SSID";
10 const char* password = "WIFI_PASSWORD";
11
12 String serverName = "http://api.open-notify.org/iss-now.json";
13
14 unsigned long lastTime = 0;
15 unsigned long timerDelay = 5000;
16
17 float myLongitude = YOUR_LONGITUDE;
18 float myLatitude = YOUR_LATITUDE;
19
20 const float maxLat = myLatitude + 5;
21 const float minLat = myLatitude - 5;
22 const float maxLong = myLongitude + 5;
23 const float minLong = myLongitude - 5;
24
25 const int redLeds[] = {D8, D7, D6, D5, D4, D3, D2};
26 const int greenLed = D1;
27 const int buzzer = D0;
28
29 void setup() {
30   Serial.begin(115200);
31   for(int i=0; i<7; i++){
32     pinMode(redLeds[i], OUTPUT);
33   }
34   pinMode(greenLed, OUTPUT);
35   pinMode(buzzer, OUTPUT);
36
37   WiFi.begin(ssid, password);

```

Fig. 3: Code snippet setting the Wi-Fi credentials

```

126   for(int i=0; i<ledsToOn; i++){
127     digitalWrite(redLeds[i], HIGH);
128   }
129   if(ledsToOn != 7){
130     for(int i=6; i>=ledsToOn; i--){
131       digitalWrite(redLeds[i], LOW);
132     }
133   }
134
135   if(distance < 0){
136     for(int i=0; i<7; i++){
137       digitalWrite(redLeds[i], LOW);
138     }
139   }
140 }
141
142 void loop() {
143   // Send an HTTP POST request depending on timerDelay
144   float longitude, latitude;
145   if ((millis() - lastTime) > timerDelay) {
146     pair<float, float> coords;
147     coords = getCoordinates();
148     longitude = coords.first;
149     latitude = coords.second;
150     float distance = getDistance(longitude, latitude);
151     showStatusOnLed(distance);
152     lastTime = millis();
153   }
154   if(minLong<longitude and longitude<maxLong and minLat<latitude and latitude<maxLat){
155     digitalWrite(greenLed, HIGH);
156     digitalWrite(buzzer, HIGH);
157   }else{
158     digitalWrite(greenLed, LOW);
159     digitalWrite(buzzer, LOW);
160   }
161 }

```

Fig. 4: Loop function getting the ISS distance

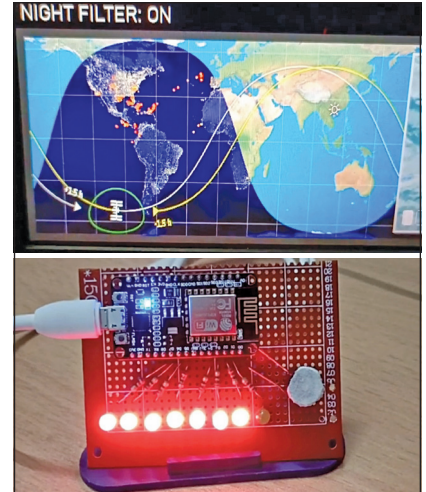


Fig. 5: Device LEDs indicating the ISS position above the device in sky

the code into NodeMCU. Next, assemble the LEDs and resistors along with NodeMCU as per circuit diagram shown in Fig. 2.

On powering the device, the LEDs will glow according to the ISS distance from the device. The ISS tracker, built using the NodeMCU platform, serves as an engaging and educational tool for monitoring the International Space Station's orbital patterns in real time.

This device offers an exciting opportunity for enthusiasts to communicate with astronauts using the ISS's radio repeater during its passes overhead. Implementing this tracker keeps users informed about the ISS's visibility, fostering a deeper appreciation for space exploration. **EFY**



Pushpendra Chandravanshi, the author, is a passionate web developer and robotics enthusiast